S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student:

Roll No.:

Semester/Year:VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation Predicate Form

Dog is an Animal isa(dog, animal)

Bird can Fly can(bird, fly)

Dog has Tail has(dog, tail)

Inheritance rule:

If

isa(X, Y) and has(Y, Z)

Then

has(X, Z)

Semantic Network

Example Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

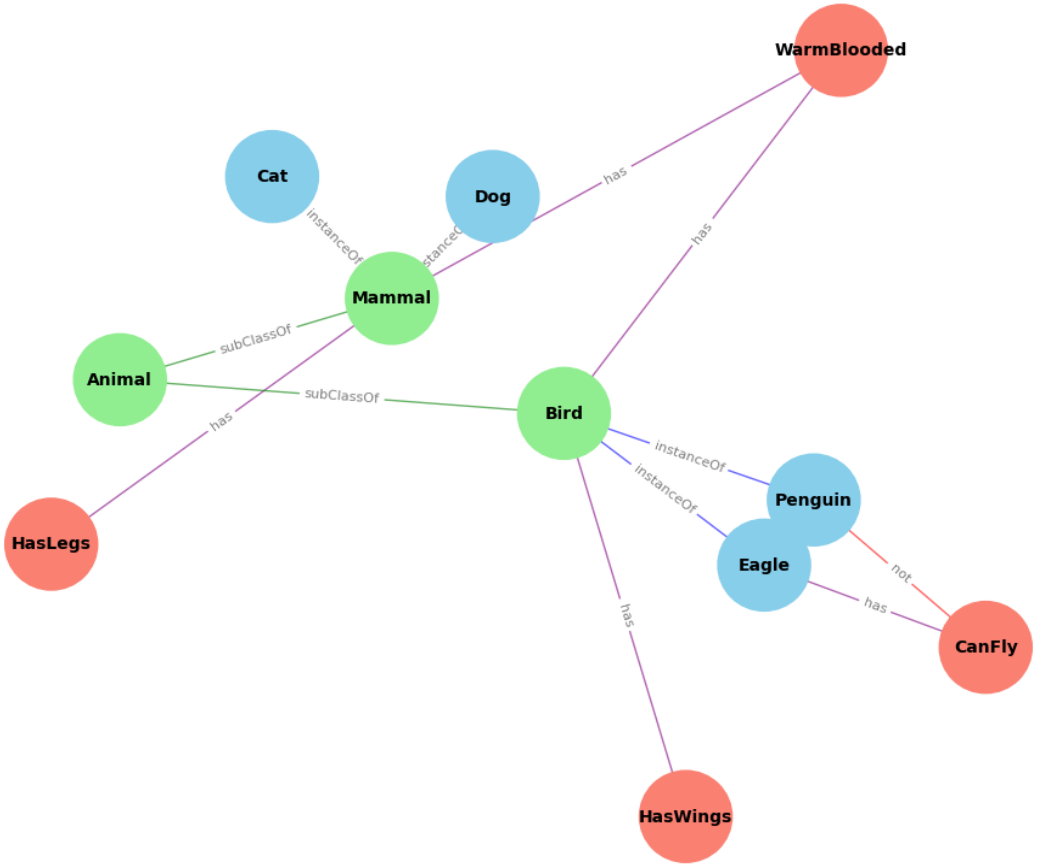
- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:

```
[1]
0s
    "Cat":      {"instanceOf": ["Mammal"]},
    "Eagle":    {"instanceOf": ["Bird"], "has": ["CanFly"]},
    "Penguin": {"instanceOf": ["Bird"], "not": ["CanFly"]},
}

def ancestors(node):
    result, queue = [], [node]
    while queue:
        n = queue.pop()
        for rel in ("instanceOf", "subClassOf"):
            for p in kb.get(n, {}).get(rel, []):
                result.append(p)
                queue.append(p)
    return result

def check(subject, predicate):
    info = kb.get(subject, {})
    if predicate in info.get("not", []): return False
    if predicate in info.get("has", []): return True
    for a in ancestors(subject):
        if predicate in kb.get(a, {}).get("not", []): return False
        if predicate in kb.get(a, {}).get("has", []): return True
    return False

# --- Test ---
tests = [
    ("Dog",     "WarmBlooded"), ("Dog",     "CanFly"),
    ("Eagle",   "CanFly"),      ("Penguin", "CanFly"),
    ("Penguin", "HasWings"),    ("Cat",     "HasLegs"),
]

for s, p in tests:
    print(f"{s:10} | {p:14} | {check(s, p)}")
```

```
Dog        | WarmBlooded    | True
Dog        | CanFly         | False
Eagle      | CanFly         | True
Penguin    | CanFly         | False
Penguin    | HasWings       | True
Cat        | HasLegs        | True
```

Knowledge Base Graph



PROGRAM CODE:

```python
isa = {
    "Bird": "Animal",
    "Dog": "Animal",
    "Sparrow": "Bird"
}
has = {
    "Animal": "Cells"
}
# CAN relationship (Ability)
can = {
    "Bird": "Fly",
    "Dog": "Bark"
}
# Function to check ISA relationship
def check_isa(x, y):
    while x in isa:
        if isa[x] == y:
            return True
        x = isa[x]
    return False
# Function to check HAS-A property with inheritance
def check_has(x, property):
    if x in has and has[x] == property:
        return True
    while x in isa:
        x = isa[x]
        if x in has and has[x] == property:
            return True
    return False
# Function to check CAN ability with inheritance
def check_can(x, ability):
    if x in can and can[x] == ability:
        return True
    while x in isa:
        x = isa[x]
        if x in can and can[x] == ability:
            return True
    return False
print("Sparrow ISA Bird :", check_isa("Sparrow", "Bird"))
print("Dog ISA Animal :", check_isa("Dog", "Animal"))
print("Sparrow HAS Cells :", check_has("Sparrow", "Cells"))
print("Bird CAN Fly :", check_can("Bird", "Fly"))
print("Dog CAN Bark :", check_can("Dog", "Bark"))
```

INPUT & OUTPUT:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

[Running] python -u "c:\Users\LAB-112\Desktop\BHAVANA BAMBAL\Practical_8_AI_BHAVANA_BAMBAL.py"
Sparrow ISA Bird : True
Dog ISA Animal : True
Sparrow HAS Cells : True
Bird CAN Fly : True
Dog CAN Bark : True

[Done] exited with code=0 in 0.058 seconds
```

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a semantic network?
2. What is ISA relationship?
3. What isinheritance in knowledge representation?
4. How is predicate logic used here?
5. What are advantages of semantic networks?

REFERENCES:

- *https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial intelligence/*
- *https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/*
- *https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/*